

Q1 - 25 July - Shift 1

If momentum [P], area [A] and time [T] are taken as fundamental quantities, then the dimensional formula for coefficient of viscosity is :

- (A) $[P A^{-1} T^0]$ (B) $[P A T^{-1}]$
(C) $[P A^{-1} T]$ (D) $[P A^{-1} T^{-1}]$

Space for your notes:

Q2 - 27 July - Shift 1

A torque meter is calibrated to reference standards of mass, length and time each with 5% accuracy. After calibration, the measured torque with this torque meter will have net accuracy of :

- (A) 15% (B) 25% (C) 75% (D) 5%

Space for your notes:

Q3 - 27 July - Shift 2

An expression of energy density is given by

$$u = \frac{\alpha}{\beta} \sin\left(\frac{\alpha x}{kt}\right), \text{ where } \alpha, \beta \text{ are constants, } x \text{ is}$$

displacement, k is Boltzmann constant and t is the temperature. The dimensions of β will be :

- (A) $[ML^2T^{-2}\theta^{-1}]$ (B) $[M^0L^2T^{-2}]$
(C) $[M^0L^0T^0]$ (D) $[M^0L^2T^0]$

Space for your notes:

Q4 - 28 July - Shift 1

The dimensions of $\left(\frac{B^2}{\mu_0}\right)$ will be :

(if μ_0 : permeability of free space and
B : magnetic field)

- (A) $[M L^2 T^{-2}]$ (B) $[M L T^{-2}]$
(C) $[M L^{-1} T^{-2}]$ (D) $[M L^2 T^{-2} A^{-1}]$

Space for your notes:

Q5 - 28 July - Shift 2

Consider the efficiency of Carnot's engine is given

by $\eta = \frac{\alpha\beta}{\sin\theta} \log_e \frac{\beta x}{kT}$, where α and β are constants.

If T is temperature, k is Boltzman constant, θ is angular displacement and x has the dimensions of length. Then, choose the **incorrect** option.

- (A) Dimensions of β is same as that of force.
(B) Dimensions of $\alpha^{-1} x$ is same as that of energy.
(C) Dimensions of $\eta^{-1} \sin\theta$ is same as that of $\alpha\beta$
(D) Dimensions of α is same as that of β

Space for your notes:

Q6 - 29 July - Shift 1

Questions

MathonGo

Given below are two statements : One is labelled as Assertion (A) and other is labelled as

Reason (R).

Assertion (A) : Time period of oscillation of a liquid drop depends on surface tension (S), if density of the liquid is ρ and radius of the drop is r ,

then $T = k \sqrt{\frac{\rho r^3}{S}} s^{3/2}$ is dimensionally correct,

where K is dimensionless.

Reason (R) : Using dimensional analysis we get R.H.S. having different dimension than that of time period.

In the light of above statements, choose the correct answer from the options given below.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

Space for your notes:

Q7 - 29 July - Shift 2

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Questions

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Match List I with List II.

List I		List II	
A.	Torque	I.	Nms^{-1}
B.	Stress	II.	J kg^{-1}
C.	Latent Heat	III.	Nm
D.	Power	IV.	Nm^{-2}

Space for your notes:

Choose the correct answer from the options given below:

(A) A-III, B-II, C-I, D-IV

(B) A-III, B-IV, C-II, D-I

(C) A-IV, B-I, C-III, D-II

(D) A-II, B-III, C-I, D-IV

Answer Key

Q1 (A)

Q2 (B)

Q3 (D)

Q4 (C)

Q5 (D)

Q6 (D)

Q7 (B)

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Q1 (A)

• Viscosity = pascal.second

$$P^x A^y T^z = [M^1 L^{-1} T^{-1}]$$

$$[M^1 L^{-1} T^{-1}]^x [L^2]^y [T^1]^z = M^1 L^{-1} T^{-1}$$

$$M^x L^{-x+2y} T^{-x+z} = M^1 L^{-1} T^{-1}$$

$$x = 1 \quad x + 2y = -1 \quad -x + z = -1$$

$$y = -1$$

$$z = 0$$

$$\text{Viscosity} = P^1 A^{-1} T^0$$

Q2 (B)

Dimensional formula for Torque

$$[\tau] = [ML^2T^{-2}]$$

Now

$$\text{Percentage error in torque} = \% \tau = \% M + 2 \% L$$

$$2 \% T$$

$$\% \tau = 25\%$$

Q3 (D)

$$\frac{\alpha[L]}{[ML^2 T^{-2}]} = [M^0 L^0 T^0]$$

$$\alpha = [ML^1 T^{-2}]$$

$$\frac{\alpha}{\beta} = \frac{[ML^2 T^{-2}]}{[L^3]} \Rightarrow \beta = \frac{[ML^1 T^{-2}][L^3]}{ML^2 T^{-2}}$$

Q4 (C)

$$u = \frac{B^2}{2\mu_0}$$

$u \rightarrow$ Energy per unit volume

$$\left[\frac{B^2}{\mu_0} \right] = [u] = \frac{[ML^2 T^{-2}]}{[L^3]} = [ML^{-1} T^{-2}]$$

Q5 (D)

$$[\alpha\beta] = [\eta] = [\sin \theta] = \text{Dimensionless}$$

$$[\eta^{-1} \sin \theta] = [\alpha\beta] = \text{D.L.}$$

Q6 (D)

$$T = k \sqrt{\frac{\rho r^3}{s^{3/2}}}$$

$$\text{Dimensions of RHS} = \frac{[M^{1/2}L^{-3/2}][L^{3/2}]}{[MT^{-2}]^{3/4}} = M^{1/8} L^0 T^{3/2}$$

Dimensions of L.H.S $\neq$ Dimensions of R.H.S

$\therefore$ option (D)

Q7 (B)

$$\text{Torque} = F \times r_{\perp} \quad \text{Nm}$$

$$\text{Stress} = \frac{\text{Force}}{\text{Area}} \quad \text{N/m}^2$$

$$\text{Latent heat} = \frac{\text{Energy}}{\text{Mass}} \quad \text{J Kg}^{-1}$$

$$\text{Power} = \frac{\text{Work}}{\text{Time}} \quad \text{N ms}^{-1}$$

A-III, B-IV, C-II, D-I